PART A [20 MARKS]

INSTRUCTION: Write you answer in answer booklet.

1. What is the comparator output for inputs shown in Figure 1?

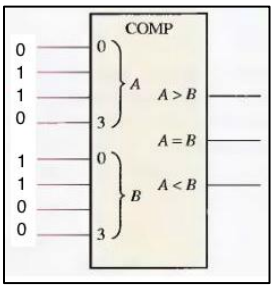


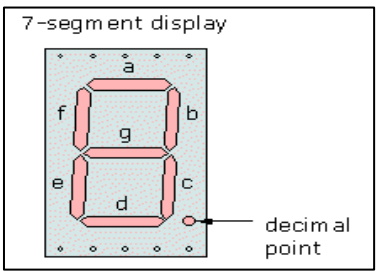
Figure 1

A. Output pin A>B is equal 1

B. Output pin A=B is equal 1

C. Output pin A<B is equal 1

D. All outputs are disabled



3. Figure 3 shows Highest Priority BCD Decoder (HPRI/BCD) IC. What is the binary output if the input pin 2, 3, 5 and 8 are pressed simultaneously?

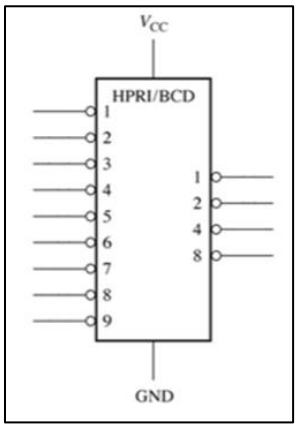


Figure 3

A. 0010
B. 0011
C. 0101
D. 1000

5. Table 1 represents the truth table for which latch?

Table 1

Input		Output
X	Y	Q
0	0	Invalid
0	1	1
1	0	0
1	1	No Change

A. Active-HIGH S-R latch
B. Active-LOW S-R latch
C. Gated S-R latch
D. Gated D latch

6. Flip-flops are normally used for the following applications, **except**:

A. Counter
B. Analog conversion
C. Frequency division
D. Data storage

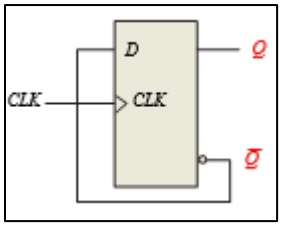


Figure 4

PART B [80 MARKS]

INSTRUCTION: Answer all questions in the answer booklet EXCEPT questions with timing diagram in Q2b. You must show all work to receive credit.

QUESTION 1 [20 MARKS]

a) Draw a schematic block diagram of three (3) bit Parallel Adder using Full Adders. Perform the following arithmetic 3-bit addition shown in Figure 6 and show all the values for this binary additional in your drawing. [7m]

$$\begin{array}{rccc} & 1 & 0 & 1 \\ + & 0 & 1 & 1 \\ \hline & & & \end{array}$$

Figure 6

b) A Bin/Dec decoder is used to activate a series of devices by a microcontroller as shown in Figure 7. Answer the following questions. [7m]

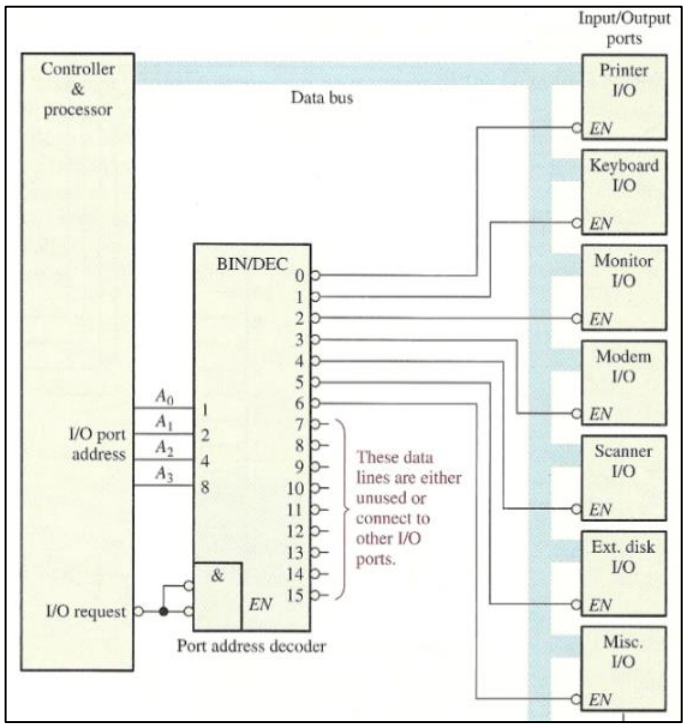


Figure 7

i) Is the Bin/Dec decoder active HIGH or active LOW? What does that mean?

ii) Which device will be activated when the input decoder is 0011?

iii) What is the input decoder when External Disk I/O is activated?

iv) If Input/output is selected as Monitor I/O, draw a simple decoder circuit using inverter and NAND gate.

c) Four different inputs from source *I* are sent to four different output destination *Y* using Mux and Demux as shown in Figure 8. Based on the input of I_3, I_2, I_1, and I_0 given in Figure 8, and inputs S_1, S_0 and M_1, M_0 in Table 3, give the binary answer for *G* and output Demux of Y_3, Y_2, Y_1, and Y_0. [6m]

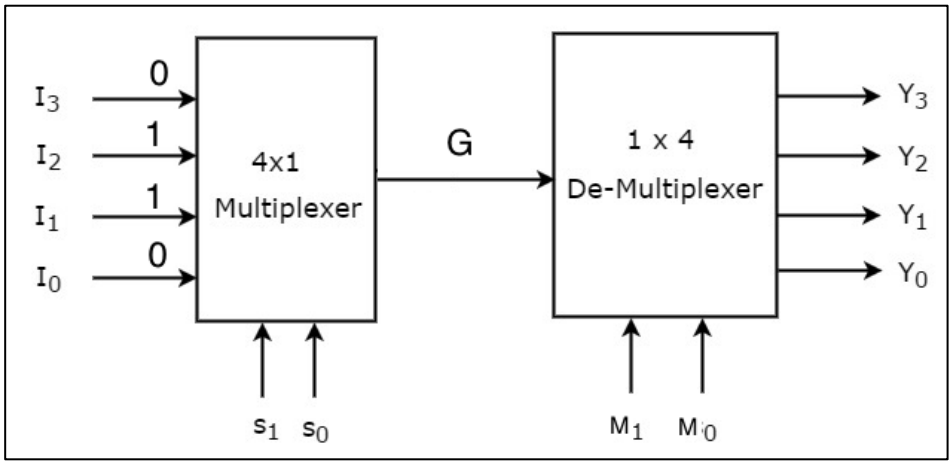


Figure 8

Table 3

S_1 S_0	G	M_1M_0	$Y_3Y_2Y_1Y_0$
01	(i)	10	(iv)
11	(ii)	01	(v)
10	(iii)	00	(vi)